Personalized Interstitial Cystitis (IC) Food App

Food and menu scanning that learns from personal tolerance



Problem Background

IC affects individuals so food guidance needs to reflect individual experience.

- Interstitial cystitis (IC), also called bladder pain syndrome, is a chronic condition causing bladder or pelvic pain. IC affects 1% of people in the United State and impacts people of all ages, races, and ethnicities. Women in particular are diagnosed with IC twice as much as men.

- Broad IC food lists can be hard to apply to packaged foods and menus, where ingredient names may be unclear.

- Tolerance of specific foods varies by person. The app records portions and personal responses rather than assuming every commonly flagged food affects everyone alike.

- **SOLUTION:** combine sourced IC guidance with personal food and symptom history, then suggest suitable alternatives at current grocery store.



User, Audience, and Context

For people with IC who want food guidance that learns from their own experience.

At a grocery store, the user scans a barcode or photographs a package and ingredient label to review a food before purchasing.

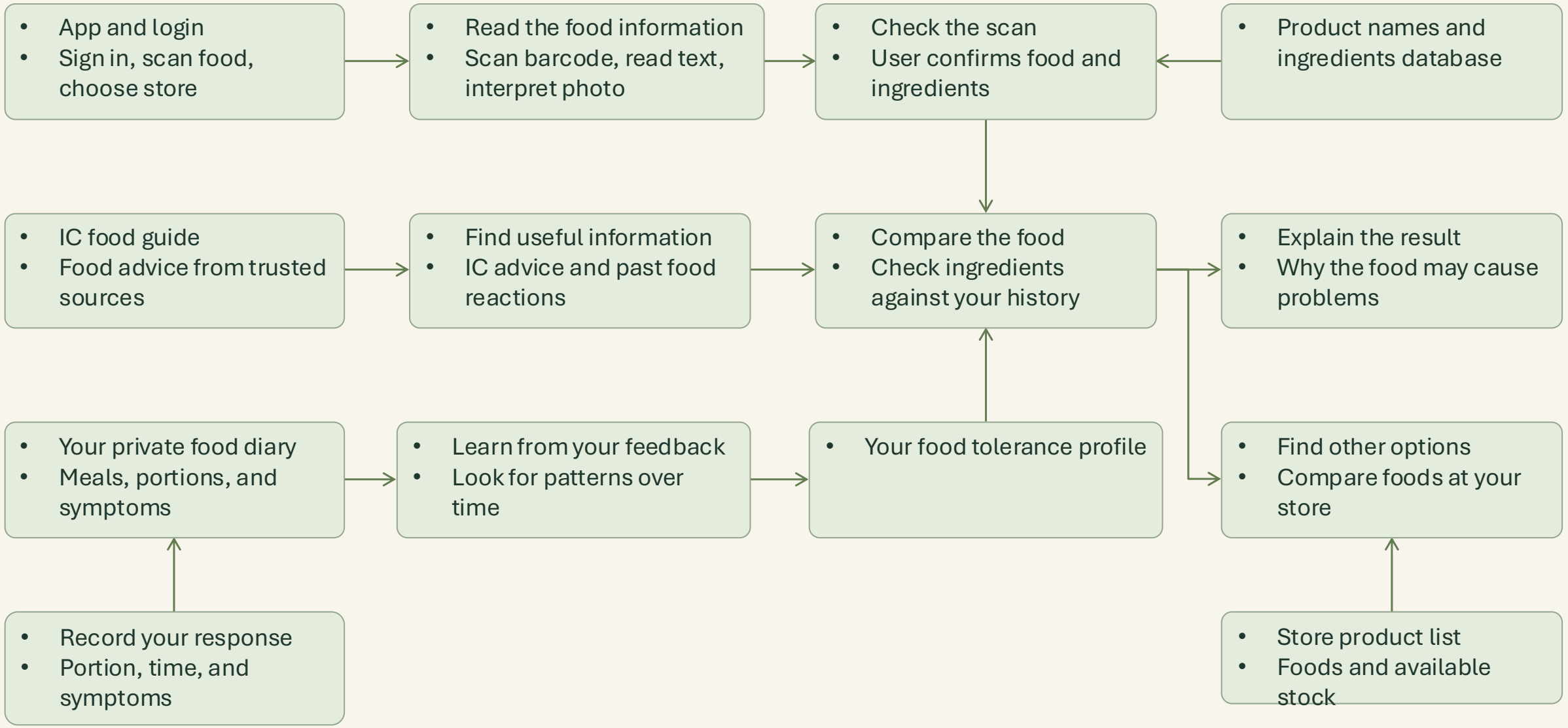
At a restaurant, the user photographs a menu or takes a picture of their plated meal.

After eating, the user records the portion, time, and how their bladder tolerated the food, including changes from their baseline symptoms.

Over time, the user receives guidance that references both general IC sources and their own previously tolerated or irritating foods.



App Structure



Public Data Availability & Class Topics

Preliminary Data Sources

Open Food Facts

Barcode lookup, ingredients, and label images. Open data with ODbL license requirements.

USDA FoodData Central

Branded food ingredients and UPC codes. Public-domain downloads or an API key for requests.

NIDDK IC dietary guidance

Public guidance for a small, source-linked knowledge base of potential dietary concerns.

IC Network food lists

Publicly readable guidance. Confirm reuse permission before incorporating the lists.

NIDDK ICDB research study

Diet and urinary-symptom data available by request. Public questionnaire can inform logging.

Methods

APIs , Pipelines and Bash Scripts

Retrieve product records and schedule refreshes. Use scraping to fill database. Normalize ingredient names; validate product, meal, portion, and symptom records.

Database + User Authentication

SQLite stores products and private user histories; login controls access to each profile.

LLMs, image recognition, and RAG

Extract label/menu text; retrieve IC guidance and past foods to explain recommendations.

Personalization + Website Design

Update ingredient scores from feedback; symptom logging, and store alternatives.

Major capabilities

Scan, explain, learn from feedback, and suggest relevant alternatives.

1. Identify potential concerns with candidate foods. Flag uncertain label recognition and incomplete menu information.

2. Remember tolerated and poorly tolerated foods, account for reported portions, and update patterns as more observations arrive.

3. Explain shared ingredients between a new food and past meals.

4. If a food is flagged as high risk, the app suggests alternatives at the current store, with availability confirmed from the store's website.

5. Log symptoms and baseline context, show recommendation confidence, correct scan errors, and let users export or delete their history.

